



**Universität
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Advanced Techniques of Machine Translation

Assignment 4

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Contents

1	True or False	3
A	False	3
B	False	3
C	True	3
2	Language Representations	3
A	Word Representation	3
B	Consequences of Increasing Vocabulary Size	3
3	Sequence-to-Sequence Models	4
A	Necessity of Encoder-Decoder Models	4
B	Non Seq2Seq Model Use Case	4
4	Open Vocabulary Translation	5
A	Sub-word Level NMT	5
B	Advantage of Using Smaller Subword Vocabulary	5
C	Disadvantage of Sharing Subwords Across 100 Languages	5
5	NMT Architectures	5
A	Gates Out Range and Function	5
B	Transformer Building Blocks	5
C	Masking	6
6	Learning from Monolingual or Multilingual Data	6
A	Monolingual Data in NMT	6
B	Motivations for Multilingual NMT	6
C	Multi-Source, Multi-Way and Massively Multilingual Models	6
7	Advanced Training Objectives	7
A	Exposure bias, Teacher forcing, Minimum Risk Training	7
B	Label Smoothing and Word-Level Knowledge Distillation	7
8	Linguistic Phenomena	8
A	Retraining Strategy	8
B	Evaluation	8
C	Context Influence	8

1 True or False

A False

Multiple good translations can make training more challenging because the model may receive conflicting information during training, as it typically uses a single reference translation per sentence. This can lead to reduced accuracy or the need for multiple references in the loss function, which complicates training. Thus, while multiple references can improve evaluation, they do not simplify training.

B False

Neural MT can learn to disambiguate homographs. Models can use surrounding context to infer the intended meaning and produce the correct translation. When context is minimal, the model may still make mistakes. So the words "cannot learn" (at all) are incorrect.

C True

Length normalization applied during decoding reduces the model's tendency toward producing short, high-probability outputs and therefore typically increases the average length of produced translations.

2 Language Representations

A Word Representation

In a word-level neural model, input words are represented as vector embeddings. Each word in the vocabulary is mapped to a fixed-size vector. These embeddings are learned during training, allowing the model to capture semantic and syntactic similarities between words. For example, "London" and "Paris" will be closer to each other than to "iPhone."

For output words, the model generates a probability distribution over the entire target vocabulary. This is typically done using a final linear layer and the softmax function that generates a probability distribution over the vocabulary. The word corresponding to the highest probability is chosen as the output.

B Consequences of Increasing Vocabulary Size

Increasing the vocabulary size has major consequences for computational complexity. Some examples are:

- Softmax Calculation: Probability distribution over the vocabulary has to account for the increased size. This leads to additional calculations and lower probability for the actual prediction as probability has to be distributed among a larger number of words.

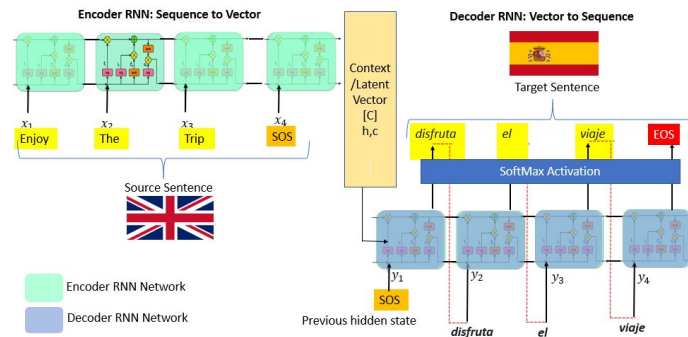


Figure 1: Encoder-Decoder Architecture; Image Source: [pluralsight.com](https://www.pluralsight.com)

- **Embedding:** The matrix used for converting words to embedding has to be larger. Algorithms like word2vec need to account for larger matrix size leading to more computational complexity.

3 Sequence-to-Sequence Models

A Necessity of Encoder-Decoder Models

A language model only generates sequences based on previous words, but for translation, we need to map from a source sequence to a target sequence. An encoder-decoder model first encodes the entire source sentence into a context vector, which the decoder then uses to generate the target sentence step by step. For example, translating "I am going to fail this course" to French "Je vais échouer dans sa course" requires understanding the whole input before generating output, which a language model alone cannot do nicely. A language model might generate text but not based on a source sentence. The following image 1 demonstrates the encoder decoder architecture.

B Non Seq2Seq Model Use Case

Sequence-to-sequence is not appropriate for tasks like text classification, where the output is a single label rather than a sequence. For example, in sentiment analysis, input is a sequence of words, but output is a sentiment class (e.g., positive, negative).

4 Open Vocabulary Translation

A Sub-word Level NMT

Subword-level NMT solves the problem of out-of-vocabulary (OOV) words by breaking words into smaller units like subwords or characters. This allows the model to handle rare or unseen words by composing them from known subwords.

B Advantage of Using Smaller Subword Vocabulary

In cases with fewer training samples, a smaller subword vocabulary helps because it reduces the risk of overfitting by having more frequent subwords, and it ensures that each subword has enough examples to learn meaningful representations. A larger vocabulary might lead to many rare subwords with poor embeddings.

C Disadvantage of Sharing Subwords Across 100 Languages

Sharing a single subword model across 100 languages can lead to several problems. Firstly, it may cause subword ambiguity, where the same subword represents different meanings in different languages, leading to poor translation quality. Second, the model may struggle with languages that have different scripts, as the shared subword vocabulary might not adequately capture the characteristics of all languages. Third, there is a risk of underrepresenting some languages, as the subword model might be dominated by languages with more samples, resulting in poor performance for the low sample ones.

5 NMT Architectures

A Gates Out Range and Function

Gates have an output range between 0 and 1, achieved using sigmoid activation function. This allows them to control how much information to forget or pass through.

B Transformer Building Blocks

The two building blocks of a transformer are:

- **Multi-Head Self-Attention:** This allows the model to look at different parts of the input sequence simultaneously and capture different types of relationships. Each "head" attends to the sequence in a unique way. The outputs from all heads are combined in the end to give the model a better understanding of the sequence compared to a single "head".
- **Position-wise Feed-Forward Network (FFN):** After attention combines information from different positions, the FFN applies the same small

neural network independently to each token representation to transform it non-linearly.

C Masking

Masking is used in the decoder to prevent the model from peeking at future words during training, ensuring that predictions are based only on previous words. If Peter removes masking, the cross-entropy might improve because the model can cheat by using future context, but during inference, when future words are not available, the model would perform poorly due to mismatch between training and testing.

6 Learning from Monolingual or Multilingual Data

A Monolingual Data in NMT

Monolingual data can be used to improve NMT by providing additional target language text to train the decoder or source language text to train the encoder. The two techniques can be explained as follows:

- **Combining MT with Autoencoding:** Here, the model is trained to reconstruct input sentences, which helps learn better representations without parallel data.
- **Backtranslation:** This involves translating monolingual target language data back to the source language to create synthetic parallel data, which is then used to train the MT system.

B Motivations for Multilingual NMT

First, it allows for zero-shot translation between language pairs not seen during training by leveraging shared representations. Second, it improves languages with fewer training samples by transferring knowledge from languages with a higher number of training samples. For example, a multilingual model trained on English-French and English-German can translate between French and German without direct parallel data.

C Multi-Source, Multi-Way and Massively Multilingual Models

- **Multi-source:** Uses multiple source languages or inputs to translate to a single target, improving translation quality by combining evidence from multiple sources.
- **Multi-way:** Translates between multiple language pairs simultaneously with shared parameters.

- **Massively Multilingual:** Involves many languages in a single model, aiming to improve languages with low samples through transfer learning.

7 Advanced Training Objectives

A Exposure bias, Teacher forcing, Minimum Risk Training

- **Exposure bias:** This occurs when during training, the model uses ground truth previous words, but during inference, it uses its own predictions, leading to error accumulation.
- **Teacher forcing:** Teacher forcing is the training strategy where the model is fed the true previous token at every step instead of its own predicted token. It makes training faster and more stable, but it also creates exposure bias, because the model never experiences its own errors during training. In other words, teacher forcing leads directly to the mismatch that causes exposure bias.
- **Minimum Risk Training:** Minimum Risk Training directly optimizes the model with respect to sequence-level evaluation metrics (e.g., BLEU), rather than token-level cross-entropy. By sampling model outputs and scoring them using the actual evaluation metric, MRT partially reduces exposure bias because the model is trained on its own predictions.

B Label Smoothing and Word-Level Knowledge Distillation

- **Label Smoothing:** This regularizes the model by replacing hard labels (0 or 1) with smoothed labels to prevent overconfidence. For example, instead of training the model to assign probability 1.0 to “forest,” label smoothing might assign 0.9 to “forest” and distribute 0.1 over all remaining words.
- **Word-level knowledge distillation:** This uses a teacher model to provide a full probability distribution over the vocabulary for each target word, and the student model learns to match that distribution. This gives the student informative “hints” about alternative plausible words that the teacher believes could fit. For instance, if the teacher predicts 0.7 “river”, 0.2 “bank”, 0.1 “cat,” the student learns a richer signal than just the gold label.

Label smoothing works on the true labels, while distillation uses teacher’s outputs.

8 Linguistic Phenomena

A Retraining Strategy

One strategy is to include gender information as an additional input to the model. For example, we could add a token indicating the speaker's gender in the source sentence. We would need to modify the training data to include such tokens and retrain the system. A disadvantage is that it requires annotated data with gender information, which might not be available.

B Evaluation

We could evaluate using automatic metrics like BLEU if we have references, but for gender agreement, human evaluation might be better to check consistency. Automatic evaluation might not capture grammatical agreement well, so for accurate and truthful results, human evaluation is recommended. For large-scale testing, automatic metrics can be used if references include gender variations.

C Context Influence

Yes, a system with more context could handle agreement better by using information from previous sentences. For example, if the previous sentence mentions "the woman", the model can infer feminine gender. We could use a context window of multiple sentences or entire documents during training to capture such dependencies.